Run 1: Engineering Document Quality Assurance Report

Project: AI-Powered Engineering Document Quality Assurance Agent | Training simulation

1. Objective

Review a fictional centrifugal pump maintenance procedure against a basic engineering-document quality checklist. Identify missing information, ambiguous instructions, and areas requiring verification. This is a simulated review, not approval for real maintenance work.

2. Source Document

Document ID: DEMO-PMP-001
Equipment: Centrifugal Pump P-101

1. Stop the pump before starting maintenance.
2. Isolate the equipment from relevant energy sources.
3. Open the pump and inspect its internal components.
4. Replace damaged components.
5. Reassemble the pump and restart it.
6. Record the maintenance activity.

Training notes: the procedure does not specify how isolation is verified; does not identify required PPE; provides no inspection acceptance criteria; and does not define restart authorization requirements.

3. Executive Summary

The six-step procedure provides a basic sequence for maintenance, but several details needed for a controlled, reviewable procedure are missing. The most significant gaps concern energy-isolation verification and restart authorization. Other gaps include PPE requirements, inspection acceptance criteria, and component replacement criteria. These are document-quality findings; they do not prove that the actual pump or an existing site procedure is unsafe. Severity ratings below are preliminary and should be validated against the site's risk assessment and approved engineering requirements.

4. Findings

ID / Severity	Evidence	Issue and potential impact	Recommended action
F-01 **High***	Step 2: “Isolate the equipment from relevant energy sources.”	The procedure does not describe how isolation is verified before maintenance. This may leave an important safety control unclear.	Reference the approved site isolation and verification procedure. Confirm requirements with qualified site personnel.
F-02 **Medium***	Steps 1–6; no PPE requirements stated.	Task-specific PPE is not identified. The applicable requirements cannot be established from this sample alone.	Reference the approved task risk assessment and applicable PPE requirements; do not invent them.
F-03 **Medium***	Step 3: “Inspect its internal components.”	No acceptance criteria are provided for deciding whether components are acceptable, repairable, or require replacement.	Reference applicable OEM documentation and approved engineering acceptance criteria.
F-04 **High***	Step 5: “Reassemble the pump and restart it.”	Required pre-start checks and authorization are not described. The sample does not establish who may authorize restart.	Reference the approved restart/commissioning procedure and site authorization process.
F-05 **Medium***	Step 4: “Replace damaged components.”	Criteria for identifying damage and selecting approved replacement components are not specified.	Reference approved parts information, OEM requirements, and authorized repair criteria.

*Severity is a preliminary reviewer judgment, not a confirmed site risk rating.

5. Checklist Summary

Checklist area	Run 1 assessment
Required steps and completeness	Partially addressed; full procedural completeness not verifiable.
Safety and isolation	Gap identified; verification method not specified.
PPE requirements	Missing from the sample.
Inspection acceptance criteria	Missing from the sample.
Restart authorization	Missing from the sample.
Evidence and traceability	Each finding references a step or document omission.
Technical claims	No numerical limits were added; external standards were not checked.
Unresolved questions	OEM requirements, site procedures, risk assessment, and equipment-specific criteria require verification.

6. Work Log and Limitations

Sources inspected: the fictional six-step procedure and the supplied quality checklist. Method: compare each step against checklist areas, record evidence, classify the issue, and recommend a verification source. Limitations: no OEM manual, approved site procedure, risk assessment, equipment configuration, or maintenance records were supplied. This report identifies documentation gaps and questions for review; it does not validate the procedure for real-world use.

7. Run 1 Baseline for Comparison

Baseline observations to compare with Run 2: five preliminary findings; explicit document-step references; general corrective recommendations; limited checklist-by-checklist traceability; and a need to make uncertainty and severity limitations more explicit. Claim that Run 2 improved only after comparing the actual Run 2 output.